

Practice problems #3

1. Determine the value(s) of h such that the matrix is the augmented matrix of a consistent linear system.

(a)

$$\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 4 & -2 \\ 3 & h & -6 \end{bmatrix}$$

2. Determine which matrices are in reduced echelon form and which others are only in echelon form.

(a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(d)

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

3. Row reduce the matrices below (this means: Apply row operations to make it into) to its reduced echelon form:

$$(a) \begin{bmatrix} 1 & 0 & 0 & 3/2 \\ 0 & 0 & 1 & -1/3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

Answers are on page 2.

#1(a): $h \neq 2$

#1(b): all real h

#2: Reduced echelon form: (a),(b)

Echelon form (d)

not in echelon form (c)

#3(a):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

#3(a):

$$\begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$